

A Branch-And-Benders-Cut approach to solve the maximum flow blocker problem

Isma Bentoumi¹, Fabio Furini², A. Ridha Mahjoub³ and Sébastien Martin⁴

Abstract—Given a directed graph with capacities and interdiction costs associated with its arcs, the maximum flow blocker problem (MFBP) asks to find a minimum-cost subset of arcs to be removed from the graph in such a way that the remaining maximum-flow value does not exceed a given threshold. The MFBP has applications in telecommunication networks and in the monitoring of civil infrastructures, among others. We propose an integer linear programming formulation (ILP) with an exponential number of constraints, called Benders cut, for the MFBP. Accordingly, we derive a branch-and-cut algorithm to optimally solve the problem. Preliminary experimental results are reported to assess performance of the formulation and more precisely to determine the dimension of the problem that could be solved to proven optimality.

I. INTRODUCTION

Let $G = (V, A)$ be a directed graph where A is a set of m arcs and V is a set of n vertices with a source $s \in V$ and a destination $t \in V$. Each arc $a \in A$ is given a *capacity* $c_a \in \mathbb{Z}_+$. We also denote an arc a as (u, v) where $u \in V$ is the tail and $v \in V$ is the head of the arc. Let y_a be a non-negative continuous variable representing the value of the flow passing through an arc $a \in A$. A *feasible flow* from the source to the destination, or simply a flow, has to respect the following two types of constraints. The first m constraints are called the *capacity constraints* for the arcs:

$$0 \leq y_a \leq c_a, \quad a \in A, \quad (1)$$

and the second set of n constraints are called the *flow-conservation constraints* for the vertices:

$$\sum_{a \in \delta^+(u)} y_a = \sum_{a \in \delta^-(u)} y_a, \quad u \in V, \quad (2)$$

where $\delta^+(u) = \{(u, v) \in A : v \in V \setminus \{u\}\}$ (the outgoing arcs) and $\delta^-(u) = \{(v, u) \in A : v \in V \setminus \{u\}\}$ (the incoming arcs). Without loss of generality, we consider graphs in which the source has only one entering arc from the destination, i.e., $\delta^-(s) = (t, s)$ and the destination has only one outgoing arc, i.e., $\delta^+(t) = (t, s)$. Finally, let $\varphi(G)$ be the function

returning the value of the flow outgoing from the source in a graph G :

$$\varphi(G) = \sum_{a \in \delta^+(s)} y_a, \quad (3)$$

which is also equal to the value of the flow entering the destination as well as the flow on the arc (t, s) , due to the flow conservation constraints (2).

One of the fundamental optimization problem consists in determining the maximum flow of a graph from the source to the destination and accordingly the maximum flow value. This problem, called, the *maximum flow problem* (MFP) has been largely studied since its introduction in the 1950s, see e.g., [4].

This paper studies the *blocker variant* of the MFP in which each arc $a \in A$ is also given an *interdiction cost* $r_a \in \mathbb{Z}_+$. This problem, called the *maximum flow blocker problem* (MFBP), consists in finding a minimum-cost subset of arcs to be interdicted, i.e., removed from the graph, in such a way that the maximum flow value between s and t in the remaining graph is no larger than a given threshold. The threshold is called the *target flow* and it is denoted by Φ .

a) Applications: The problem is motivated by applications in several domains. In telecommunication networks, the flow routed on an arc represents the amount of data sent from the source to the destination and passing through that arc. However, some arcs may be malfunctioning due to anomalies, failures or packet loss caused by congestion. In this context, MFBP optimal solutions allow analyzing the network resilience in case of malfunctioning arcs, modeled by removing the arc from the graph. Precisely, considering a network with blocker costs equal to one, it is said to be resilient to f simultaneous failures if after the removal of any set of at most f arcs, there exists a flow of value larger than the target flow Φ . Accordingly, an optimal MFBP solution is a set of arcs of minimum cardinality $\zeta(G)$ such that the network is not resilient to $\zeta(G)$ simultaneous arc failures with respect to a target flow Φ . By considering $f = \zeta(G) - 1$, since $\zeta(G)$ is the minimum size of the set of arcs, removing any set of arcs of size at most $\zeta(G) - 1$ (maximum simultaneous malfunctioning arcs) ensures that the maximum flow value in the remaining graph is larger than Φ .

A further application of the MFBP is in the context of monitoring civil infrastructure, where the aim is to monitor the transportation links in order to reduce illegal traffics .

b) Related work: To the best of our knowledge, there are no articles dealing with the MFBP. However, a closely related problem, called the *maximum flow interdiction problem*

*This work was not supported by any organization

¹ Isma Bentoumi is with LAMSADE UMR CNRS 7243 University Paris-Dauphine PSL. Place du Marechal de Lattre de Tassigny 75016 Paris France. isma.bentoumi@dauphine.eu

² Fabio Furini is with Department of Computer Control and Management Engineering “Antonio Ruberti”. Sapienza University of Rome. Via Ariosto 25. 00185 Roma. Italy. fabio.furini@uniroma1.it

³ A. Ridha Mahjoub is with LAMSADE UMR CNRS 7243 University Paris-Dauphine PSL. Place du Marechal de Lattre de Tassigny 75016 Paris France. ridha.mahjoub@lamsade.dauphine.fr

⁴ Sébastien Martin is with Huawei Technologies France. 19 Quai du Point du Jour 92100 Boulogne-Billancourt France. sebastien.martin@huawei.com

(MFIP), has been largely studied in the literature. Given a graph, together with capacities and interdiction costs on the arcs and an *interdiction budget* denoted by Ψ , the MFIP consists in finding a subset of arcs of total interdiction cost no larger than Ψ to be removed from the graph in such a way that the maximum flow value is minimized. The MFIP is the closest problem to the MFBP since they share a similar bilevel structure, see [3] for a survey on bilevel optimization. It has been shown to be $\mathcal{NP}$ -hard in [5]. Moreover, remark that the MFIP and the MFBP have the same decision problem that consists of finding a set of interdicted arcs such that the maximum flow value remaining in the non-interdicted graph is less than a given threshold and the total interdiction cost does not exceed a limited budget. Which leads to the following result given by Proposition 1.

Proposition 1: The MFBP is $\mathcal{NP}$ -hard.

In [5], the first bilevel model for the MFIP is proposed as well as a technique to obtaining a single-level formulation. In [6], a generic Benders decomposition algorithm is presented to solve bilevel interdiction problems.

c) Our contribution: The aim of this paper is to propose the first Integer Linear Programming (ILP) formulation to solve the MFBP to proven optimality. After having introduced the bilevel formulation of the MFBP in Section II, we derive in Section III an ILP formulation for the MFBP. This formulation has m variables associated to the arcs of the graph and an exponential number of constraints. Accordingly, we have developed a Branch-and-Cut algorithm that is described in Section IV. Finally, Section V reports the preliminary experimental results we obtain by solving the ILP formulation with the Branch-and-Cut algorithm proposed.

II. A BILEVEL FORMULATION FOR THE MFBP

In this section, we design a bilevel optimization model to solve the MFBP. This model is based on two types of variables, the first-level decisions variables associated to the so-called *leader problem* which affect the second-level decision variables associated to the so-called *follower problem*. For the MFBP, the leader determines a set of interdicted arcs to be removed from the graph and the follower determines the maximum flow value in the remaining graph. The leader anticipates the optimal follower's solution with the goal of choosing the minimum cost subset of acts to be interdicted that results in a remaining graph having a maximum flow value no larger than Φ .

Let us introduce a vector $\mathbf{w}$ of m binary first-level decision variables, each variable $w_a \in \{0, 1\}$ is associated to an arc $a \in A$ and it takes value 1 if and only if the arc a is interdicted, i.e., removed from the graph G . Moreover, let us introduce a second vector $\mathbf{y}$ of m non-negative continuous second-level decision variables, each variable $y_a \geq 0$ is associated to an arc $a \in A$ and it represents the value of the flow on the arc. Therefore, a bilevel model for the MFBP reads as follows:

$$\zeta(G) = \min_{\mathbf{w} \in \{0,1\}^m} \sum_{a \in A} r_a w_a \quad (4a)$$

$$\vartheta(\mathbf{w}) \leq \Phi, \quad (4b)$$

$$\text{where } \vartheta(\mathbf{w}) = \max_{\mathbf{y} \in \mathbb{R}_+^m} \sum_{a \in \delta^+(s)} y_a \quad (4c)$$

$$\sum_{a \in \delta^+(u)} y_a - \sum_{a \in \delta^-(u)} y_a = 0, \quad \forall u \in V, \quad (4d)$$

$$y_a \leq c_a (1 - w_a), \quad \forall a \in A. \quad (4e)$$

The link between the leader problem ((4a)-(4b)) and the follower problem ((4c)-(4e)) is established by the *value function* $\vartheta(\mathbf{w})$ which returns the maximum flow value in the graph induced by non-interdicted arcs. This value is determined by the follower objective function (4c) which also corresponds to the maximum value of the function $\varphi(s)$, see equation (3), in a graph where the capacity of interdicted arcs is set to 0. Constraints (4d) are the flow conservation constraints, see (2), of the vertices. Constraints (4e) model the capacity constraints of the arcs, see (1), and they impose, at the same time, a flow of value 0 on interdicted arcs. As far as the leader problem is concerned, the objective function (4a) minimises the total cost of the interdicted arcs and constraint (4b) imposes that the maximum flow value $\vartheta(\mathbf{w})$ is no larger than Φ .

Note that a binary realisation $\tilde{\mathbf{w}}$ of the first-level variables, is called an *interdiction strategy* and it generates a *non-interdicted graph* $\mathcal{G}(\tilde{\mathbf{w}}) = (V, \mathcal{A}(\tilde{\mathbf{w}}))$, i.e., the graph induced by the set of not-interdicted arcs which is denoted by $\mathcal{A}(\tilde{\mathbf{w}})$. In Theorem 2 of [2], an important structural property of the follower problem is established. In particular, given an interdiction policy $\tilde{\mathbf{w}}$, it is proven that the follower problem can be restated as follows:

$$\begin{aligned} \vartheta(\tilde{\mathbf{w}}) = \max_{\mathbf{y} \in \mathbb{R}_+^m} \Big\{ & \sum_{a \in \delta^+(s)} y_a - \sum_{a \in A} \tilde{w}_a y_a : \\ & \sum_{a \in \delta^+(u)} y_a - \sum_{a \in \delta^-(u)} y_a = 0, \quad \forall u \in V, \\ & y_a \leq c_a, \quad \forall a \in A \Big\}. \end{aligned} \quad (5)$$

In this LP reformulation, constraints (4d) are replaced with the “standard” capacity constraints (1) for the arcs. The constraints of the follower do not depend anymore on the first-level variables and a penalization term is added to the new objective function to ensure that $\vartheta(\tilde{\mathbf{w}})$ is the maximum flow value in the non-interdicted graph $\mathcal{G}(\tilde{\mathbf{w}})$.

III. A SINGLE-LEVEL ILP MODEL FOR THE MFBP

In this section, we introduce the first ILP formulation for the MFBP using only the vector $\mathbf{w}$ of binary variables, i.e., the natural variables associated with the arcs whose binary realisations represent the interdiction policies. The ILP model is obtained starting from the bilevel model (4) and projecting out the variables $\mathbf{y}$ of the follower model (5). The polytope of the feasible solutions for the follower subproblem, which

does not depend on the leader variables as shown in (5), can be defined as follows:

$$\mathcal{P}_f = \left\{ \mathbf{y} \in \mathbb{R}_+^m : \sum_{a \in \delta^+(s)} y_a \geq \Phi + 1, \right. \\ \left. \sum_{a \in \delta^+(u)} y_a - \sum_{a \in \delta^-(u)} y_a = 0, \forall u \in V \right. \\ \left. y_a \leq c_a, \forall a \in A \right\}. \quad (6)$$

We remark that only flows of value strictly larger than Φ , i.e., greater or equal to $\Phi + 1$ are considered, since all the capacities of the arcs are integer.

For this reason, the polytope $\mathcal{P}_f$ includes the constraint imposing the lower bound on the value of the flow. Model (5) is valid for any (fractional) vector $\mathbf{w} \in \mathbb{R}_+^m$ and, since the objective function is linear, it is sufficient to optimize over the set of extreme points of $\mathcal{P}_f$ which are denoted by $\text{ext}(\mathcal{P}_f)$. Constraints (4b) can be then restated as follows:

$$\vartheta(\mathbf{w}) = \max_{\hat{\mathbf{y}} \in \text{ext}(\mathcal{P}_f)} \left\{ \sum_{a \in \delta^+(s)} \hat{y}_a - \sum_{a \in A} w_a \hat{y}_a \right\} \leq \Phi. \quad (7)$$

Applying a Benders-like decomposition to the bilevel model (4), we obtain the following single-level ILP reformulation for the MFBP:

$$\zeta(G) = \min_{\mathbf{w} \in \{0,1\}^m} \sum_{a \in A} r_a w_a \quad (8a)$$

$$\sum_{a \in \delta^+(s)} \hat{y}_a - \sum_{a \in A} w_a \hat{y}_a \leq \Phi, \quad \forall \hat{\mathbf{y}} \in \text{ext}(\mathcal{P}_f). \quad (8b)$$

where constraints (4b) are replaced with the constraints (8b), called *Benders cuts*. This ILP model is called natural, since it features only the variables associated to the arcs and it is denoted by n-ILP in the remainder of the article.

IV. A BRANCH-AND-CUT APPROACH FOR THE MFBP

In order to solve n-ILP, we develop a branch-and-Benders-cut approach, i.e., a branch-and-cut (B&C) algorithm where Benders cuts (8b) are separated in the nodes of the branching tree. This exact algorithm requires to define a *relaxed master problem* (RMP) where the binary variables are replaced with continuous variables taking values between 0 and 1. In the initialization phase only a subset of constraints are included in the RMP. To check that RMP solutions respect all the Benders cuts or to determine one or more violated constraints which are then added to the RMP, we propose the following separation procedure.

Given a (fractional) solution $\mathbf{w}^*$ of the RMP in a B&C node, the *separation problem* for the Benders cuts (8b) requires finding a vector $\hat{\mathbf{y}} \in \text{ext}(\mathcal{P}_f)$ such that:

$$\sum_{a \in \delta^+(s)} \hat{y}_a - \sum_{a \in A} w_a^* \hat{y}_a > \Phi \quad (9)$$

or to prove that such a vector does not exist, i.e., that all Benders cuts are satisfied by the solution $\mathbf{w}^*$. The inequality (9) can be equivalently rewritten as follows:

$$\sum_{a \in \delta^+(s)} (w_a^* - 1) \hat{y}_a + \sum_{a \in A \setminus \delta^+(s)} w_a^* \hat{y}_a < -\Phi \quad (10)$$

and accordingly, it is necessary to find a vector $\hat{\mathbf{y}} \in \text{ext}(\mathcal{P}_f)$ leading to the minimum value of the left-hand-side of (10). This can be done by solving a *minimum cost circulation problem* (MCCP) as shown in the next proposition which characterizes the complexity of the *separation problem* for the Benders cuts (8b):

Proposition 2: The *separation problem* for the Benders cuts (8b) can be solved in polynomial time for fractional and integer solutions $\mathbf{w}^*$ of the RMP.

Proof: Let us introduce a vector $\mathbf{p} \in \mathbb{R}^m$ where each element corresponds to the flow cost p_a of an arc $a \in A$ and a vector $\mathbf{l} \in \mathbb{R}^m$ where each element corresponds to the lower bound l_a on the flow value passing through an arc $a \in A$. The MCCP aims to find a minimum cost flow respecting the capacity constraints (1), the flow conservation constraints (2) and the lower bounds imposed on the flow of the arcs, i.e., the constraints of the polytope $\mathcal{P}_f$, where $l_{(t,s)} = \Phi + 1$ and $l_a = 0$ for every other arcs $a \in A \setminus (t, s)$. By setting $p_a = (w_a^* - 1)$ for the arcs $a \in \delta^+(s)$ and $p_a = w_a^*$ for the arcs $a \in A \setminus \delta^+(s)$, if the optimal solution value of this MCCP is strictly smaller than $-\Phi$, then the Benders cuts of maximum violation is found. Otherwise all Benders cuts are satisfied by solution $\mathbf{w}^*$. Since the MCCP can be solved in polynomial time by the exact algorithms for the *Minimum Cost Flow Problem* (see [1]), the separation problem of the Benders cuts (8b) can be solved in polynomial time for any (fractional) solution $\mathbf{w}^*$. ■

V. EXPERIMENTAL RESULTS

In this section, we assess the computational performance of n-ILP (8). Our primary aim is to investigate the strength of this formulation and the dimension of the problem that could be solved to proven optimality. We used CPLEX 12.7.0 (called CPLEX for brevity) and the CALLABLE LIBRARIES to implement our branch-and-cut algorithm. It is worth noticing that in the initialization phase of the algorithm, only one Benders cut is added to the RMP. It is obtained by solving a MFP in the graph, which, by definition, provides a flow of value greater than Φ .

Two important parameters are investigated for this study. The first parameter is the size of the instances, characterized by the number of vertices n and the arc-density of the graph $d(G) = |A|/(n^2 - n)$. The second parameter is the target flow Φ defined as a percentage $k \in [0, 1]$ of the maximum flow value in G . For this study, we consider a set of 180 instances generated using a random graph generator that we have implemented in Python 3.8.10. These instances represent Erdős-Rényi random $G(n, p)$ graphs where p , set

to value 0.5, is the probability for including each arc. We consider a set of instances with 3 values of n , i.e. the number of vertices, $n = 40$, $n = 60$ and $n = 100$ and four values of arc-densities, $d(G) = 0.2$, $d(G) = 0.4$, $d(G) = 0.6$ and $d(G) = 0.8$. Finally, we consider three values of k to define the target flow, $k = 0.2$, $k = 0.6$ and $k = 0.9$. For each value of n and $d(G)$, we create 5 distinct Erdős-Rényi graphs using different random seed generators. The source and the destination are randomly selected between the vertices of the graphs such that s and t are linked by at least three consecutive arcs. All arcs capacities and interdiction costs take integer values in the range $[1, 49]$, except for the arcs outgoing the source s or entering the destination t that have infinite capacities and interdiction costs. For our experiments, we consider a time limit of 600 seconds.

In Figure 1 and Table I, we discuss the performance of n-ILP on the instances described previously, depending on values of n , $d(G)$ and k .

In Figure 1, we give the computing time boxplot of n-ILP depending on k . We show in this graph, the time spent by the natural formulation through the quartiles. The lines extending vertically indicate the variability outside the upper and lower quartiles. The y-axis is the computing time (in logarithmic scale) and the x-axis represents values of k . On the top part of the figure, we report for each value of k , the total number of instances solved to proven optimality ($\#opt$). This boxplot shows that performance of the formulation increases with k . Precisely, for $k = 0.2$, 28 instances have been solved to proven optimality, while 53 instances have been optimally solved for $k = 0.9$. This is due to the fact that the number of Benders cut decreases when the target flow increases.

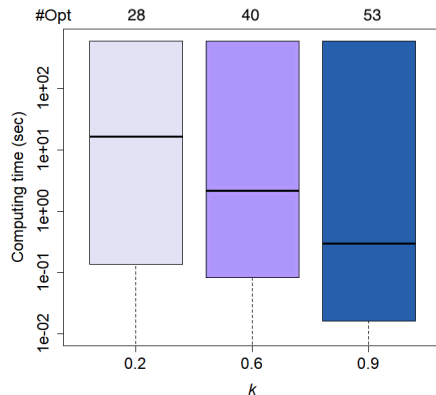


Fig. 1: Computing time boxplot of n-ILP, depending on k .

In Table I, we report, for each group of graphs (same number of vertices n and same density $d(G)$), the total number of instances considered ($\#$), the number of instances solved to proven optimality ($\#opt$), the average computing time in seconds and the average number of Benders cuts (8b) generated ($\#cuts$). We observe increasing computing time for larger graphs, which is consistent with the fact that the number of constraints increases with the size of the graph. For example, for graphs of 40 vertices with a density $d(G) = 0.8$,

TABLE I: Performance of n-ILP, depending on the number of vertices n and the density $d(G)$.

n	$d(G)$	#	#opt	time	#cuts
40	0.2	15	15	0.1	8
	0.4	15	15	4.8	85
	0.6	15	15	13	135
	0.8	15	14	46.7	232
60	0.2	15	15	0.1	34
	0.4	15	13	81.2	191
	0.6	15	7	321.7	1396
	0.8	15	6	364.3	1460
100	0.2	15	9	264.4	231
	0.4	15	5	403.4	398
	0.6	15	5	430.1	986
	0.8	15	2	534	1198
Total		180	121		

one instance has not been solved to proven optimality within the time limit, while for graphs of 100 vertices, even for $d(G) = 0.2$, only 9 instances have been optimally solved.

VI. CONCLUSIONS

In this paper, we have studied the maximum flow blocker problem. We proposed a natural formulation with an exponential number of constraints, called Benders cuts. Accordingly, we developed a Branch-and-cut algorithm for the MFBP. As a future line of research, one may actually wonder whether there exists valid inequalities to strengthen the natural formulation or if another efficient procedure can be investigated to solve the MFBP.

ACKNOWLEDGMENT

The authors are grateful to Ivana Ljubic for illuminating discussions on the subject which also helped us in the presentation of this paper.

REFERENCES

- [1] R. K. Ahuja, T. L. Magnanti, and J. B. Orlin. *Network Flows: Theory, Algorithms, and Applications*. Prentice hall, N.Y, 1993.
- [2] K. J. Cormican, D. P. Morton, and R. K. Wood. Stochastic network interdiction. *Operations Research*, 46(2):184–197, 1998.
- [3] S. Dempe. *Bilevel Optimization: Theory, Algorithms, Applications and a Bibliography*, chapter 20, pages 581–672. Springer International Publishing, 2020.
- [4] A. Schrijver. On the history of the transportation and maximum flow problems. *Mathematical Programming*, 91(3):437–445, 2002.
- [5] R. K. Wood. Deterministic network interdiction. *Mathematical and Computer Modeling*, 17(2):1–18, 1993.
- [6] R. K. Wood. Bilevel network interdiction models: Formulations and solutions. *Wiley Encyclopedia of Operations Research and Management Science*, 174:1–11, 2011.